One of the things I love about cosmology is how much it requires thinking creatively, trying to approach the physics of the universe from a totally new direction. This doesn't mean fully unconstrained flights of fancy. You can't just randomly make stuff up. But what you can (and must) do is constantly find new ways to look at problems to wring a little more insight out of whatever data the universe has to offer.

This kind of creative thinking becomes especially important when we're faced with a conundrum like "How do we improve on Concordance Cosmology or the Standard Model?" Everything we've tried so far has been frustratingly consistent with predictions; where are we supposed to find clues leading us to new models if we can't get something in the current model to break?

Clifford Johnson is optimistic, and points out that this lack of clear direction might be good for us. "I don't have a thing I can point to and go, 'This is the future!'" he told me. "I just feel the diversity of things that we've been driven to do … is probably somewhat healthy."

So, we're branching out. There are radio surveys attempting to illuminate the cosmic Dark Ages between the time of the CMB and the epoch of the first stars, in the hope that some departure from Concordance Cosmology might reveal itself more starkly. There are new kinds of gravitational wave detectors that rely on techniques as different as quantum interference between atoms and combining signals from pulsars. These might, in an indirect way, bring us information about the behavior of black holes or the physics of the early universe. Experiments looking at new ways to find dark matter might show us how to expand the Standard Model of particle physics, or shift our thinking in cosmology. Studies of the polarization of the CMB could show us signatures of cosmic inflation that completely change our understanding of the early universe. Or, the lack of such signals could motivate more studies into inflation alternatives like bouncing cosmologies. Laboratory experiments studying alternative ideas about the energy of the vacuum might finally solve the problem of dark energy, if it's not a cosmological constant after all. It may even be possible, with observations spanning decades, to measure the expansion of the universe *directly* by staring at a distant source for so long that its apparent speed away from us changes.

Pedro Ferreira is also optimistic about this diversity of approaches. "I think it all might look quite specialized and bitty," he says, but having a